Rapid Microstructure Prediction in Laser Powder Bed Fusion Using Bayesian Updating of Eagar–Tsai Model

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Core Thesis

Composition

- A 5-element affine projection represents the accessible composition simplex.
- Interior coloring encodes local element content across the design space.
- Bayesian updates narrow the search to feasible LPBF compositions.

Solidification Condition

- Process parameters determine thermal gradients, cooling rates, and melt-pool shape.
- The Eagar–Tsai model provides fast temperature-field priors across the parameter space.
- These fields define the local solidification path for each candidate build condition.

Solidified Microstructure

- Microstructure emerges from the coupling of chemistry and thermal history.
- Predicted G/R conditions map to expected grain morphology and phase selection.
- The workflow targets rapid microstructure prediction from composition to final structure.

Motivation and Constraints

- LPBF alloy design spans large composition–processing spaces.
- Resulting microstructures depend on thermal gradients, G, and solidification rates, R.
- Fast thermal models are needed for high-throughput screening before expensive simulations or experiments.

Key constraint

The Eagar–Tsai model is fast enough for screening, but its simplified physics limits fidelity:

- temperature-invariant properties,
- analytical melt-pool assumptions,
- reduced accuracy relative to FEM thermal simulations.

Reusable Two-Column Layout

Stage or category A

- Use this side for requirements, assumptions, or process details.
- Keep bullets short enough to read during a live talk.

Stage or category B

- Mirror the structure on the left when comparison matters.
- Use `\deliverable{}` and `\risk{}` for emphasis.

Downselection rule

Reject options that fail the practical constraints, even if their isolated performance looks strong.



Gradient or Tradeoff Visual

Example design gradient

High-temperature region

Structural region

Replace this annotation with the key takeaway for the figure.

Refractored ET Model

Core idea

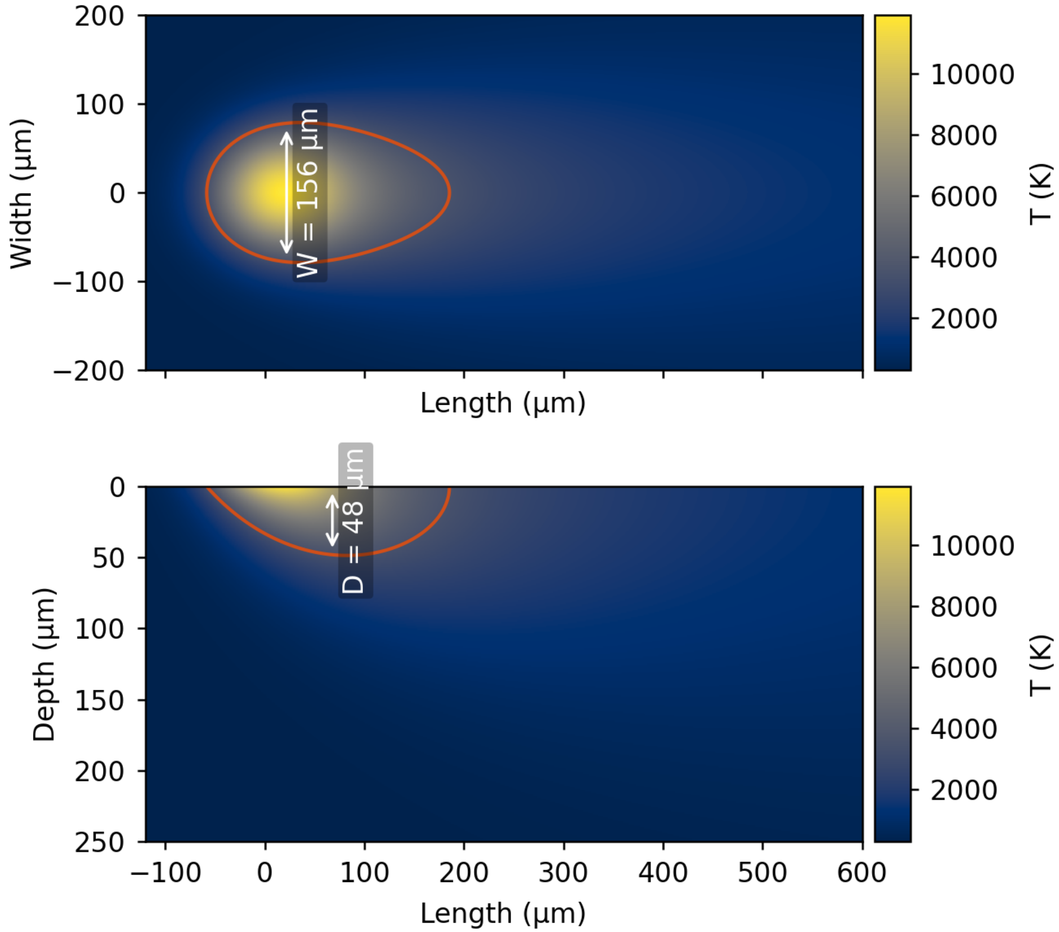
The refactored Eagar–Tsai workflow separates melt-pool prediction, 2D heatmap generation, and 3D temperature-field export into reusable steps.

- Library-backed execution supports direct melt-pool evaluation from process and material inputs.
- Temperature fields can be exported as both dense CSV volumes and VTI files for downstream analysis.
- The workflow is structured for Bayesian correction, microstructure inference, and repeatable parameter studies.

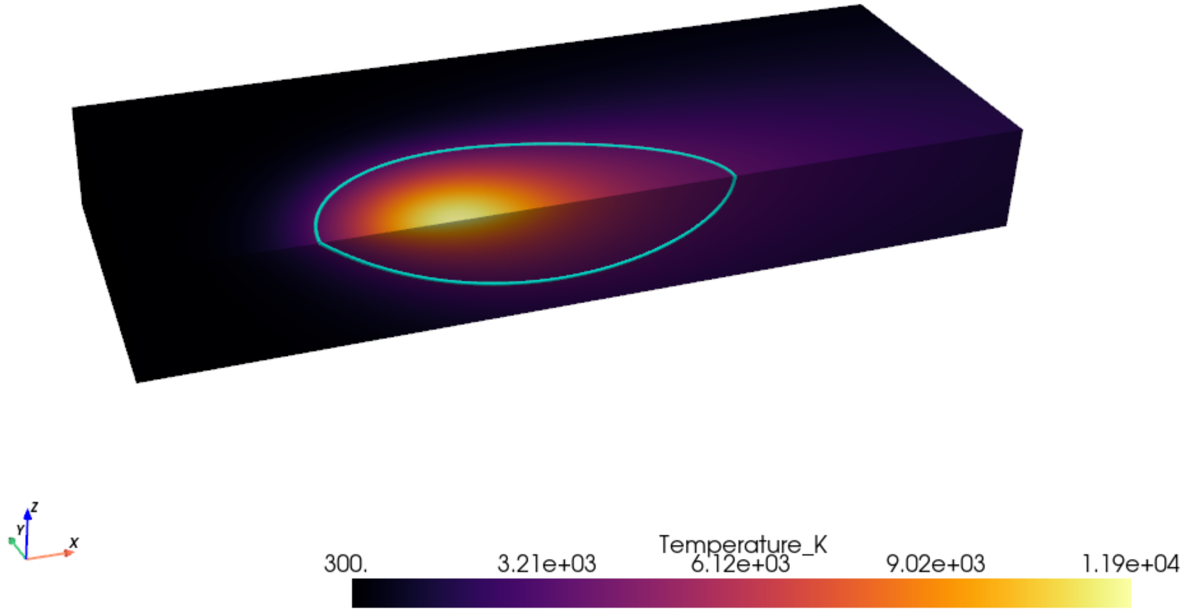
Note to self: add speedup results and refer to Dogu’s JOS paper.



ET Thermal Cross Sections

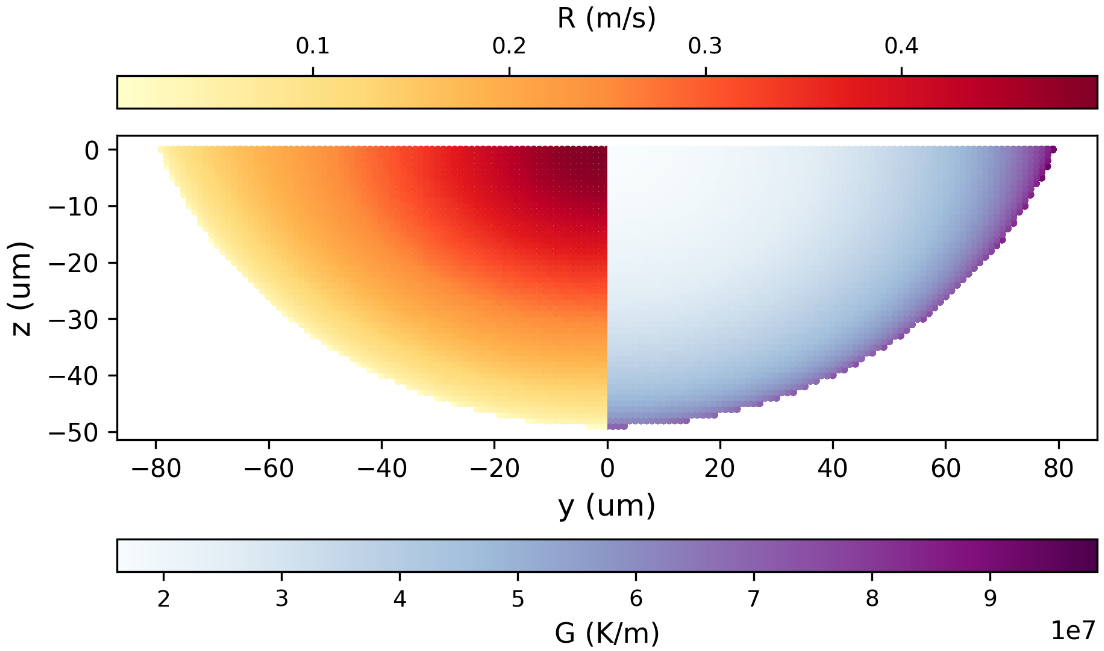


Temperature heatmap for $Hf_2Mo_2Ta_{48}W_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s. Top: $z = 0$ surface. Bottom: $y = 0$ cross section.



3D view of the ET melt pool for the same alloy and process condition.

ET Thermal Gradient and Solidification Rate



Projection of G and R along the solidification front onto a yz slice for $Hf_2Mo_2Ta_{48}W_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.

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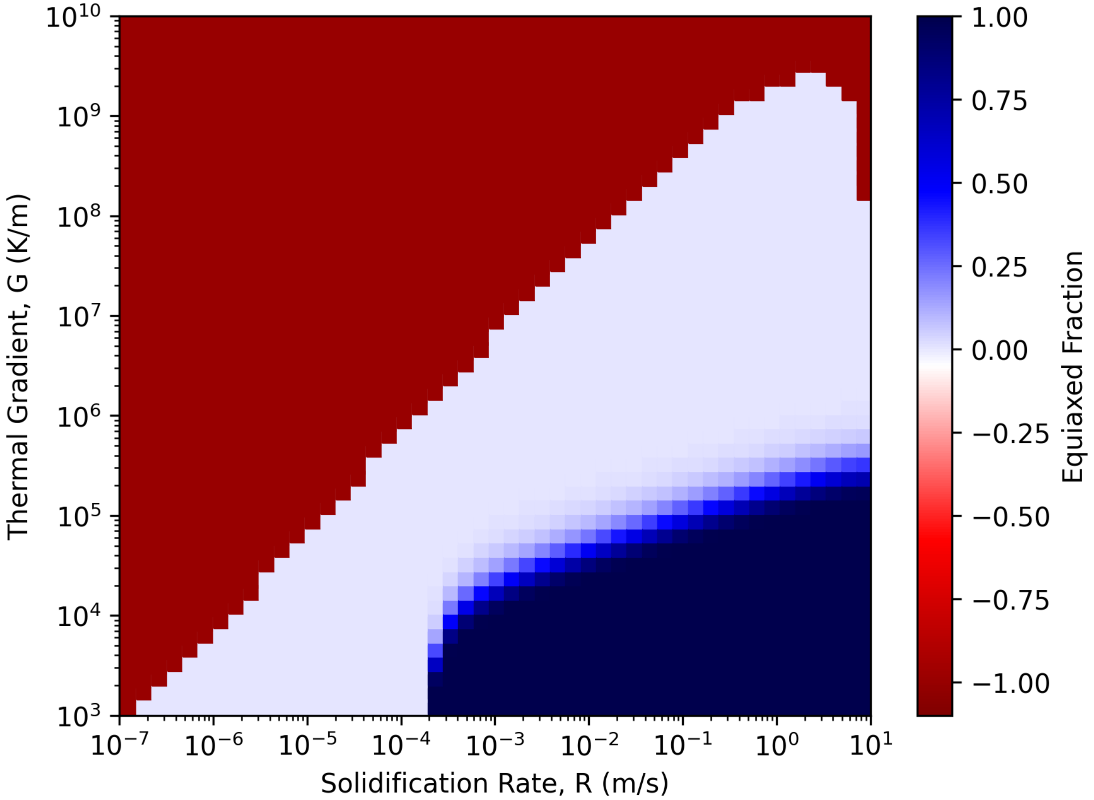
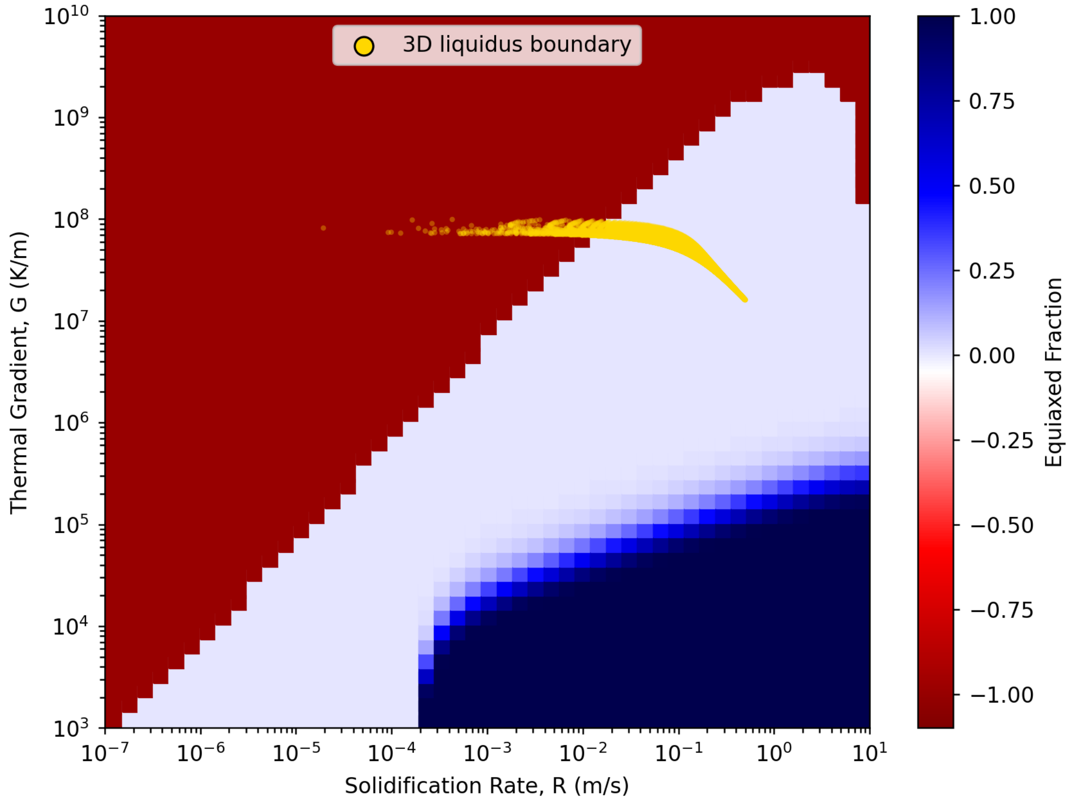
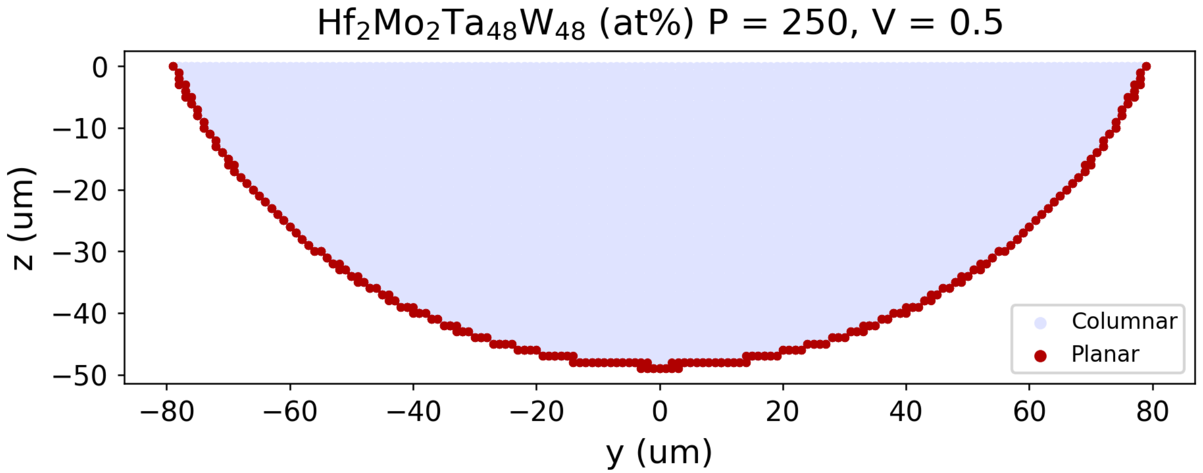


GR Map calculated with Thermo-Calc CET Property Model for $Hf_2Mo_2Ta_{48}W_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.

ET Microstructure Classification



GR Map calculated with Thermo-Calc CET Property Model for $Hf_2Mo_2Ta_{48}W_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.



Projected microstructure classification along the solidification front onto a yz slice for $Hf_2Mo_2Ta_{48}W_{48}$ (at.%), $P = 250$ W and $V = 0.5$ m/s.

Thermo-Calc AM Module

TCAM Thermal Cross Sections

TCAM Thermal Gradient and Solidification Rate

TCAM Microstructure Classification

Closing

Takeaway

End with the decision, ask, or technical conclusion you want the audience to remember.

- **Deliverable**: Name the concrete output.
- **Primary risk**: Name the hardest unresolved issue.
- **Immediate next step**: Name the action that starts now.